

Session 16 Quiz - AI That Hears & Speaks - Voice AI

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:57450]

The three steps in the Voice AI recipe are Hear, _____, and Speak.

- A. Translate
- B. Process
- C. See
- D. Record

Correct Answer: B. Process

Explanation: The middle step "Process" involves the AI figuring out the meaning of the words it just heard. This is analogous to the "Think" stage in other AI recipes. Translate, See, and Record are not the correct middle step.

Q2 [Type: MCQ | Difficulty: Easy | QID:57451]

A microphone catches vibrations from your voice and turns them into a visual representation called a _____.

- A. Waveform
- B. Pixel map
- C. Sound cloud
- D. Vibration log

Correct Answer: A. Waveform

Explanation: A waveform is the "squiggly line" that represents sound vibrations in a visual format the computer can analyze. Sound cloud is a music service, pixel map is for images, and vibration log is not a standard term.

Q3 [Type: MCQ | Difficulty: Easy | QID:57453]

The technology that turns spoken audio into written words is known as _____.

- A. Voice Synthesis
- B. Audio Translation
- C. Text-to-Speech
- D. Speech-to-Text

Correct Answer: D. Speech-to-Text

Explanation: Speech-to-Text (STT, also called ASR) decodes the sound waveform into text format. Text-to-Speech does the opposite, Voice Synthesis is a broader term, and Audio Translation is not standard.

Q4 [Type: MCQ | Difficulty: Easy-Medium | QID:57454]

Compared to deep voices, high-pitched voices produce _____ and faster sound waves.

- A. Tighter
- B. Wider
- C. Lighter
- D. Higher

Correct Answer: A. Tighter

Explanation: High-pitched sounds vibrate more quickly, resulting in waves that appear more compressed or "tighter." Wider waves correspond to lower pitches. Higher, wider, and lighter do not describe the wave compression accurately.

Q5 [Type: MCQ | Difficulty: Easy | QID:57455]

When an AI reads an audiobook out loud, it is using a process called _____.

- A. Voice Capturing
- B. Text-to-Speech
- C. Automatic Recognition
- D. Natural Language Reading

Correct Answer: B. Text-to-Speech

Explanation: Text-to-Speech (TTS) converts digital text into spoken audio. Voice capturing is recording, automatic recognition is STT, and natural language reading is not a standard term.

Q6 [Type: True/False | Difficulty: Easy-Medium | QID:57456]

True or False: Voice AI is only capable of performing tasks in one language at a time.

- A. True
- B. False

Correct Answer: B. False

Explanation: Voice AI can be trained in many languages and can even be used for real-time translation and dubbing. It is not limited to a single language.

Q7 [Type: True/False | Difficulty: Easy | QID:57457]

True or False: Every individual has a unique sound waveform for their voice.

A. False

B. True

Correct Answer: B. True

Explanation: Because of different pitches, tones, and resonance, every person's voice creates a one-of-a-kind wave pattern. This uniqueness is what enables voice recognition and voice cloning.

Q8 [Type: True/False | Difficulty: Easy | QID:57459]

True or False: If an AI struggles to understand you, it might be because the audio quality is poor or there is background noise.

A. True

B. False

Correct Answer: A. True

Explanation: Distractions like noise and low-quality microphones make it much harder for the Speech-to-Text engine to decode the waveform, just as it is harder for humans to hear in a loud room.

Q9 [Type: True/False | Difficulty: Easy-Medium | QID:57460]

True or False: AI has mastered the ability to understand sarcasm as well as a human can.

A. False

B. True

Correct Answer: A. False

Explanation: Sarcasm remains a difficult challenge for AI because it relies on complex social context, tone of voice, and cultural cues. AI has not mastered this to human level.

Q10 [Type: True/False | Difficulty: Easy-Medium | QID:57461]

True or False: Voice cloning technology can be used to help individuals who have lost their physical ability to speak.

A. True

B. False

Correct Answer: A. True

Explanation: This is considered one of the "cool" or helpful sides of voice technology, providing accessibility for people who have lost their voice due to illness or injury.

Q11 [Type: MCQ | Difficulty: Medium | QID:57462]

In the Hear → Process → Speak recipe, which part relies most heavily on NLP (Natural Language Processing)?

- A. Capture
- B. Process
- C. Speak
- D. Hear

Correct Answer: B. Process

Explanation: The "Process" step is the "brain" phase where the AI analyzes the text to determine the intent and details of the request. This requires Natural Language Understanding (NLU), which is part of NLP.

Q12 [Type: MCQ | Difficulty: Easy-Medium | QID:57464]

What is the primary reason an AI might understand a certain accent better than another?

- A. The AI was trained using more data from that specific accent.
- B. The AI chooses which accents it prefers based on its algorithm.
- C. Accents with higher pitches are always easier for AI to process.
- D. Some accents are naturally easier for computers to hear.

Correct Answer: A. The AI was trained using more data from that specific accent.

Explanation: AI performance depends heavily on the diversity and volume of training data. If an accent is better represented in the training set, the AI will understand it more accurately.

Q13 [Type: MCQ | Difficulty: Medium | QID:57465]

If you receive a phone call from someone who sounds like a family member asking for urgent money, what is the safest action to take?

- A. Ask them personal questions that only they would know.
- B. Hang up and call them back at a known, trusted number.
- C. Send the money immediately since it sounds like them.
- D. Wait for them to call you back a second time to be sure.

Correct Answer: B. Hang up and call them back at a known, trusted number.

Explanation: This verifies if the caller is actually who they claim to be or if they are using a cloned voice. Calling back on a known number is the safest rule to prevent voice cloning scams.

Q14 [Type: MCQ | Difficulty: Easy-Medium | QID:57466]

Which of the following is an example of Voice AI performing an 'Accessibility' job?

- A. Using an AI voice to narrate a popular fantasy novel.
- B. Using Alexa to set a kitchen timer.
- C. Using a screen reader to read web pages aloud for a blind person.
- D. Using dictation to send a text message quickly.

Correct Answer: C. Using a screen reader to read web pages aloud for a blind person.

Explanation: This uses Text-to-Speech to provide information to users who cannot see the screen, directly addressing a disability. The other options are convenience or entertainment, not accessibility.

Q15 [Type: MCQ | Difficulty: Easy-Medium | QID:57468]

What happens during the 'STT' (Speech-to-Text) phase of Voice AI?

- A. The AI analyzes the sound waveform and decodes it into written text.
- B. The AI translates the words into a different language.
- C. The AI identifies the location of the person speaking.
- D. The AI generates a vocal response to a user's question.

Correct Answer: A. The AI analyzes the sound waveform and decodes it into written text.

Explanation: STT converts spoken audio into text, allowing the computer's logic systems to "read" what was spoken. The other options describe location tracking, translation, or TTS.

Q16 [Type: MCQ | Difficulty: Easy-Medium | QID:57469]

Which Voice AI 'Job' is occurring when a bank's automated system asks, 'How can I help you today?'

- A. Accessibility
- B. Voice Cloning
- C. Phone Bot
- D. Voice Typing

Correct Answer: C. Phone Bot

Explanation: Phone bots are AI systems specifically designed to handle customer service calls without a human agent. Accessibility is for disabilities, voice cloning mimics voices, and voice typing is dictation.

Q17 [Type: MCQ | Difficulty: Medium | QID:57470]

Why is Voice AI considered 'hard' when it comes to training data?

- A. Computers have a hard time storing the large files required for sound waves.
- B. Voice AI requires humans to manually type out every single possible sound wave.
- C. Microphones cannot pick up voices from certain parts of the world.
- D. It works worse for voices and languages that aren't well-represented in the training data.

Correct Answer: D. It works worse for voices and languages that aren't well-represented in the training data.

Explanation: If the AI has not heard enough examples of a specific voice type, accent, or language, it cannot process it accurately. This is a data diversity challenge.

Q18 [Type: MCQ | Difficulty: Easy-Medium | QID:57471]

What does a wider, slower sound wave usually indicate about the voice being captured?

- A. The voice is deep or low-pitched.
- B. The person is speaking very quietly.
- C. The voice is high-pitched and excited.
- D. The audio quality is very poor.

Correct Answer: A. The voice is deep or low-pitched.

Explanation: Lower frequencies have longer wavelengths, which appear wider and slower on a waveform. Higher pitches produce tighter, faster waves.

Q19 [Type: MCQ | Difficulty: Easy-Medium | QID:57473]

In the lesson's activity, which tool was used to design a voice by simply describing it?

- A. Siri Designer
- B. ElevenLabs
- C. Google Voice Studio
- D. Waveform Creator

Correct Answer: B. ElevenLabs

Explanation: ElevenLabs is a real tool mentioned in the lesson that allows users to generate and speak with custom-designed AI voices by describing the desired characteristics.

Q20 [Type: MCQ | Difficulty: Medium | QID:57474]

When an AI 'Process' step determines that a person said 'Set an alarm for 7am,' what specific 'detail' is it identifying?

- A. Open clock app
- B. Tomorrow
- C. Set alarm
- D. 7am

Correct Answer: D. 7am

Explanation: The detail is the specific parameter (the time) needed to complete the requested action. "Set alarm" is the action verb, but "7am" is the detail that makes the command complete.